CIBC Client Transaction Data Cleaning & Risk Analysis

Data Cleaning & Risk Detection using Alteryx

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Tool Used: Alteryx Designer
Dataset: Simulated CIBC Banking Transactions (2025)

Business Problem

Financial institutions process thousands of transactions daily. Raw transactional datasets often contain **duplicate records, inconsistent values, and missing information**, which can lead to inaccurate reporting and poor risk visibility.

The objective of this project was to:

- Clean and standardize raw banking transaction data
- Detect duplicate or invalid records
- Identify high-value transactions that may require additional review
- Generate summarized insights for business reporting

Data Quality Issues Identified

After importing the dataset into Alteryx, several issues were identified:

- Duplicate transaction records
- Null or empty rows in key fields
- Inconsistent spelling in the Region column (e.g., *toronto, Tronto, Tornoto*)
- Raw dataset containing 1,937 records with noisy data

These issues needed to be resolved before performing any analysis.

Data Cleaning Methodology

The following Alteryx tools were used to clean and prepare the data:

Select Tool

Used to modify data types and ensure fields were properly formatted.
Examples:

- Transaction_ID → Integer / String
- Amount → Double
- Transaction_Date → Date

Filter Tool

The Filter Tool was used to remove invalid rows by identifying records containing **null values in critical fields**.

Unique Tool

The Unique Tool was used to separate unique and duplicate transactions.

Duplicate rows were removed from the final dataset to maintain data integrity.

Browse Tool

The Browse Tool was used throughout the workflow to inspect the data and identify issues such as spelling inconsistencies and incorrect values.

This allowed manual validation before proceeding to further analysis.

Formula Tool

The Formula Tool was used to standardize region names by creating a new column Clean_Region.

After verification, the Clean_Region column replaced the original Region column.

Risk Flag Implementation

To identify potentially high-value transactions, a column for **High/Normal Value Flag** was created using the Formula Tool.

Logic used:

- If **Amount > 10,000** → **"High"**
- Else → "Normal"

This helped isolate transactions that may require additional financial or fraud analysis.

Data Analysis & Aggregation

Several **Summarize tools** were used to generate business insights.

Region Summary

- Total transactions by Region
- Total transaction value by Region
- Count of high-value transactions

Payment Method Analysis

- Total transaction value by Payment Method
- Percentage of high-value transactions per method

Risk Summary

- Total normal vs high value transactions

These summaries help provide insights into regional activity, payment trends, and potential financial risk patterns.

Data Export

Outputs were generated from the workflow:

Cleaned Dataset

Exported as an Excel file containing fully cleaned transaction data ready for analysis.

Analytical Report Dataset

Other Excel Files were:

- Region Summary
- Payment Method Analysis
- Risk Summary

This structured output allows business teams to quickly review key insights.

Key Insights

From the analysis:

- The dataset was reduced from 1937 to 1800 clean records
- 90 duplicate transactions were identified and removed
- 90 high-value transactions were detected using the risk flag

- Region standardization improved accuracy of regional reporting

These insights demonstrate how data cleaning directly improves data reliability and risk visibility.

Conclusion

This project demonstrates the importance of **data cleaning and automation in financial analytics**.

By building an Alteryx workflow, raw banking transaction data was successfully transformed into a clean, standardized, and analysis-ready dataset.

Through this project I gained practical experience with:

- Alteryx data preparation workflows
- Data cleaning techniques
- Risk flag creation using formulas
- Aggregation and summarization for business insights
- Exporting automated analytical reports

This workflow can be reused for future datasets, making the process scalable and efficient for real-world financial data analysis.